

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 17

Clinical Imaging of the Lower Limb

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Recommended Readings

- Drake, Vogl, Mitchell. Grays anatomy for students. Newest ed. Elsevier Churchill Livingstone.
 - Pay special attention to the green **“In the Clinic”** boxes for the clinical correlations for this topic

Objectives

SOM.MK.II.BPM1.1.MSK.1.ANAT.1336	Identify the structures of the hip joint, femur, knee joint, tibia, fibula, ankle joint and foot in imaging studies use this information to interpret clinical signs and symptoms.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1337	Describe the posterior dislocation of the hip and the nerve most likely to be affected by this injury.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1338	Identify posterior dislocation of the hip joint on a radiograph.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1297	Describe the clinical significance of an intracapsular versus an extracapsular fracture of the femur and identify on imaging studies.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1496	Identify femoral fractures in imaging studies and discuss likely clinical signs as symptoms as a result of these fractures
SOM.MK.II.BPM1.1.MSK.1.ANAT.1497	Identify injuries to the ligaments and cartilages of the knee on imaging studies
SOM.MK.II.BPM1.1.MSK.1.ANAT.1339	Describe the patellar tendon reflex including the nerve roots being tested
SOM.MK.II.BPM1.1.MSK.1.ANAT.1340	Describe a “Baker’s cyst” and explain possible sequelae.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1498	Identify fractures tibial and fibular fractures in imaging studies and discuss likely clinical signs as symptoms as a result of these fractures
SOM.MK.II.BPM1.1.MSK.1.ANAT.1324	Describe the consequences and clinical presentation of “a compartment syndrome”
SOM.MK.II.BPM1.1.MSK.1.ANAT.1499	Describe signs and symptoms resulting from various areas of blockage of the blood supply of the lower limb
SOM.MK.II.BPM1.1.MSK.1.ANAT.1500	Identify the structures of the gluteal region, compartments of the thigh and leg in imaging studies and use this information to interpret clinical signs and symptoms.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1335	Describe the development of varicose veins.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1341	Describe the Achilles tendon reflex including the nerve roots being tested.

Case 1

A 65-year-old man, (Gerald), comes to the emergency department because he experienced a severe, spasmodic pain in the right leg and thigh while walking from the car park to the shopping mall. The pain was relieved when he paused for a short rest. He has a past medical history of Diabetes Type II, Hypertension and Hyperlipidemia.

Physical examination shows the skin of his right leg is pale, cool, shiny, and hairless. Sensory testing of the anterior thigh and leg shows no abnormalities. The patellar and Achilles reflexes were normal.

Patellar tendon 'knee jerk' reflex

- Patient should be relaxed
- Strike the patella tendon
- Observe for knee extension
- Testing **L3-L4** spinal nerves



Achilles tendon reflex

- Patient should be relaxed
- Strike the Achilles tendon
- Observe for plantar flexion
- Testing the **S1-S2** spinal nerves



Case 1 cont'd

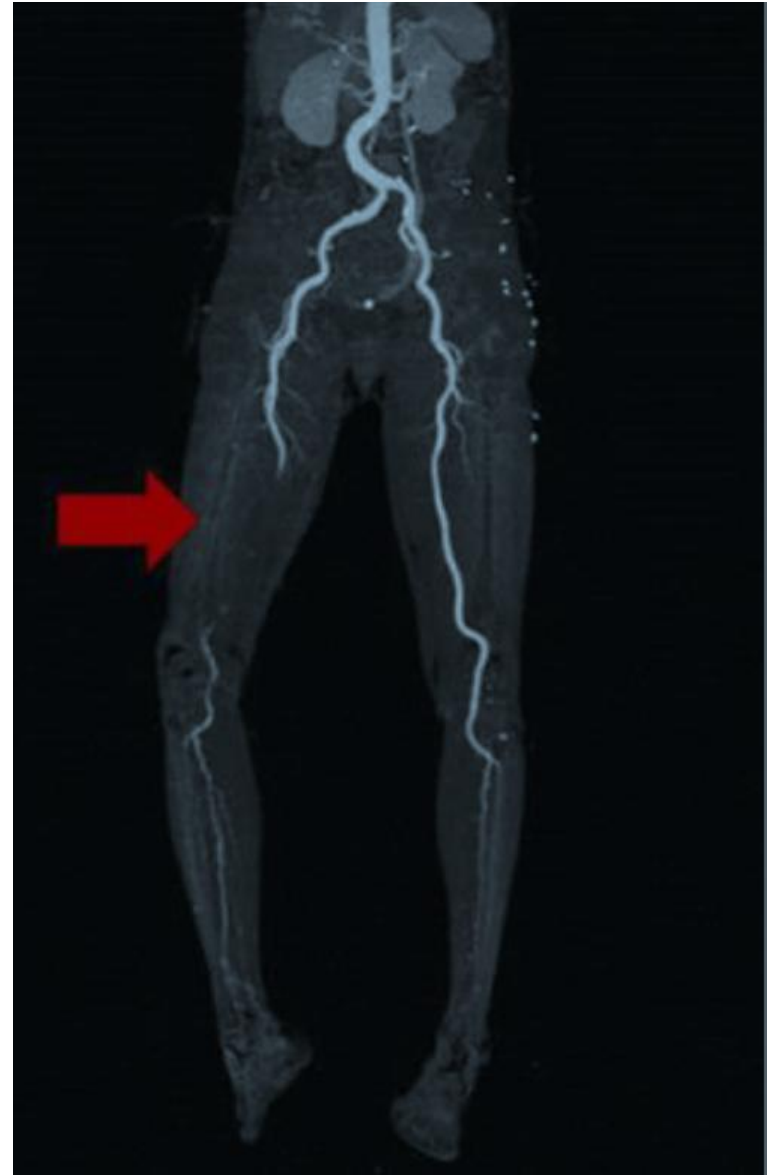
He has weakened popliteal, posterior tibial and dorsalis pedis pulses on the right side compared to the left side. The femoral pulses on both sides are equal.



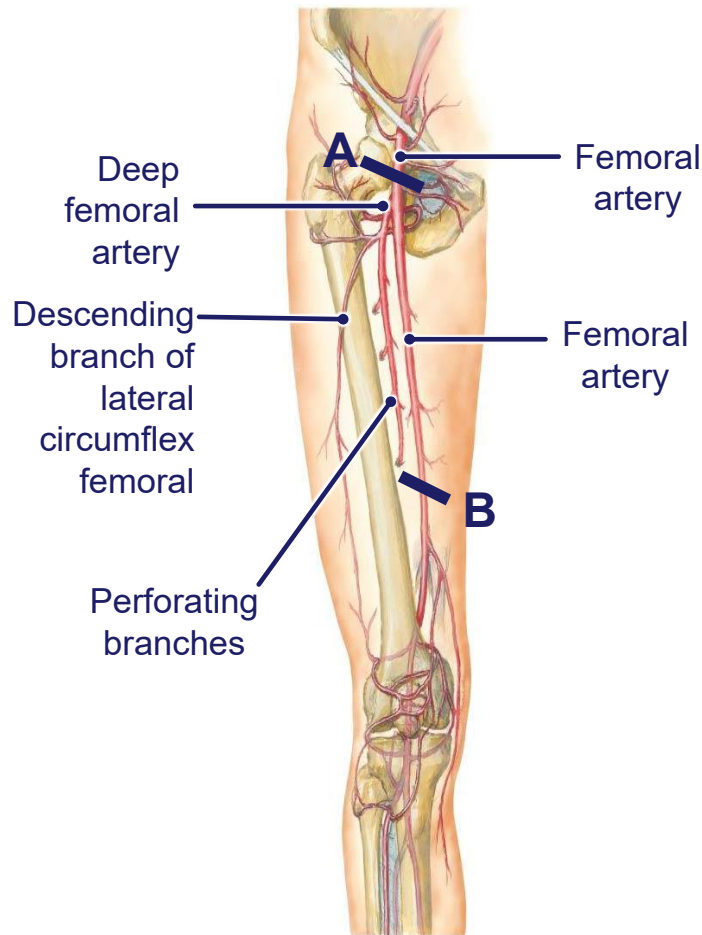
Where is the blockage?

Intermittent Claudication

- Blockage of arteries results in ischemia of the muscles in the compartment.
- This produces a cramp-like pain with exertion.
- Patient was forced to stop walking after a limited distance because of the intensity of the pain.
- With rest the pain disappears.
- However, on resumption of walking the pain re-occurs.



Blockage of the Femoral Artery



Lesion A

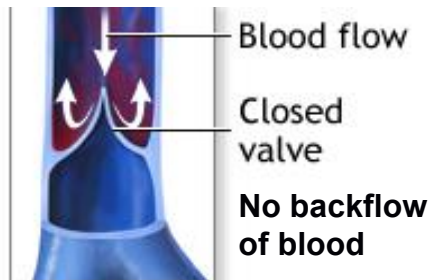
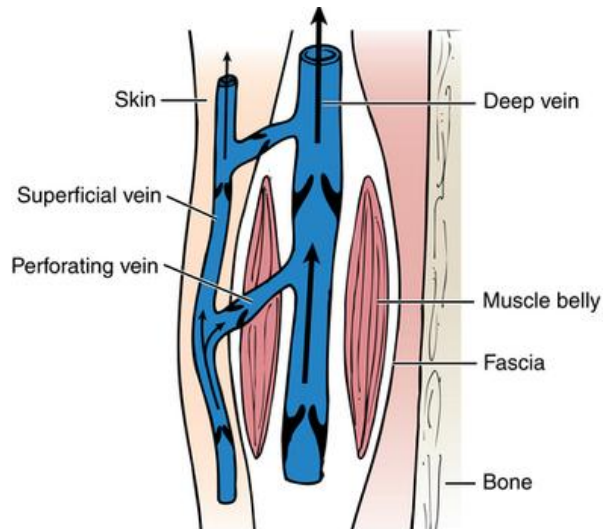
- Blockage of the proximal part of the femoral artery.
- Can be bypassed by the **cruciate and trochanteric** anastomosis.

Lesion B

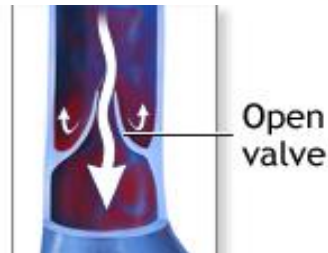
- Blockage of the distal part of femoral artery in the adductor canal.
- Can be bypassed by **descending branch of lateral femoral circumflex anastomosing with the genicular and muscular branches of the popliteal arteries**

Varicose veins

- Insufficient valves in superficial veins and perforating branches leads to blood flow from deep veins (higher pressure) to superficial veins (lower pressure).
- Continuous excess flow through from deep to superficial veins increases the pressure in the superficial veins
- This increasing pressure causes blood to pool in the superficial veins resulting in dilation of these veins – varicose veins
- Varicose veins can be treated surgically with *venous stripping of these superficial veins – what nerve(s) can be damaged due to this procedure?



Normal venous return



Allows backflow of blood



Insufficient venous return in varicose veins

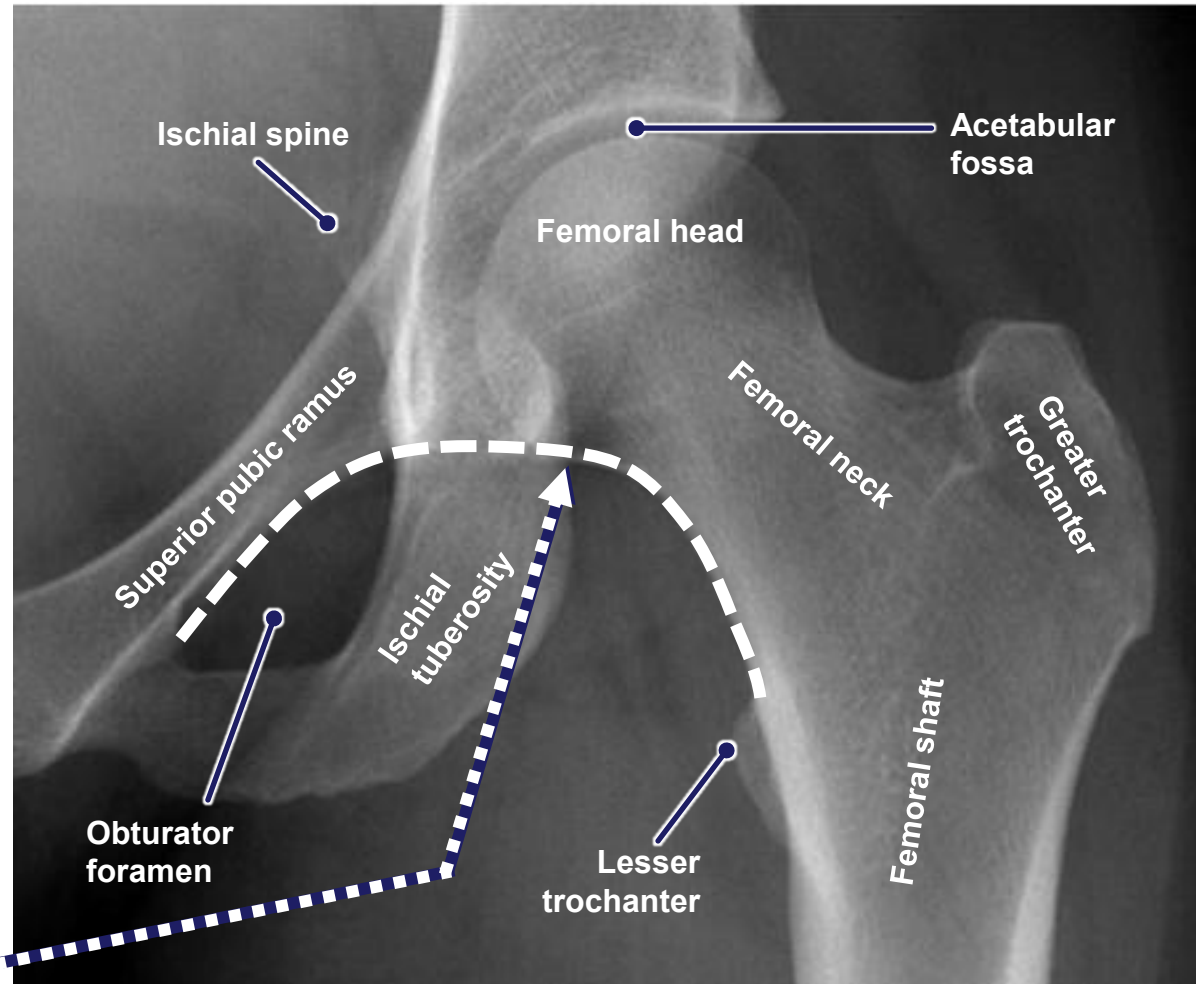


Case 2

An 84-year-old man is brought to the emergency department in the early hours of the morning after falling down several stairs. He is an active, independent, gentleman with good premorbid mobility and no significant co-morbidities. The patient has severe pain in his hip joints and obvious deformities of his lower limbs. What is the most likely injury of the hip joint that will be seen on imaging studies?

AP Radiograph of Hip Joint

1. Ischial tuberosity
2. Ischial spine
3. Acetabulum
4. Femoral head
5. Femoral neck
6. Femoral shaft
7. Greater trochanter
8. Lesser trochanter
9. Superior pubic ramus



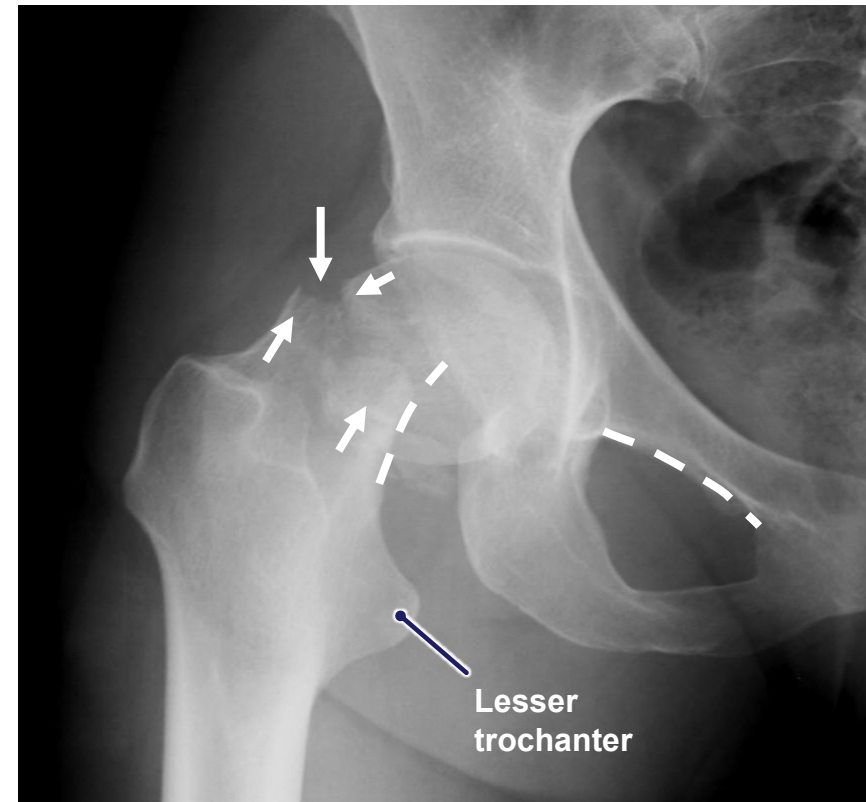
Shenton's line:

Formed by the medial edge of femoral neck
and the inferior edge of the superior pubic ramus

Proximal Femoral Fractures

Intra-capsular – Femoral Neck Fracture

- Disruption of the cortical bone indicates the presence of a fracture
- Shenton's line is disrupted
- Increased x-ray density where displaced bones overlap
- Lesser trochanter more prominent due to the external rotation
- Damages the retinacular arteries
- Compromises blood supply to the head of femur
- Risk of avascular necrosis
- Displaced intracapsular fractures (as shown in the image) can result in external rotation of the leg with some shortening

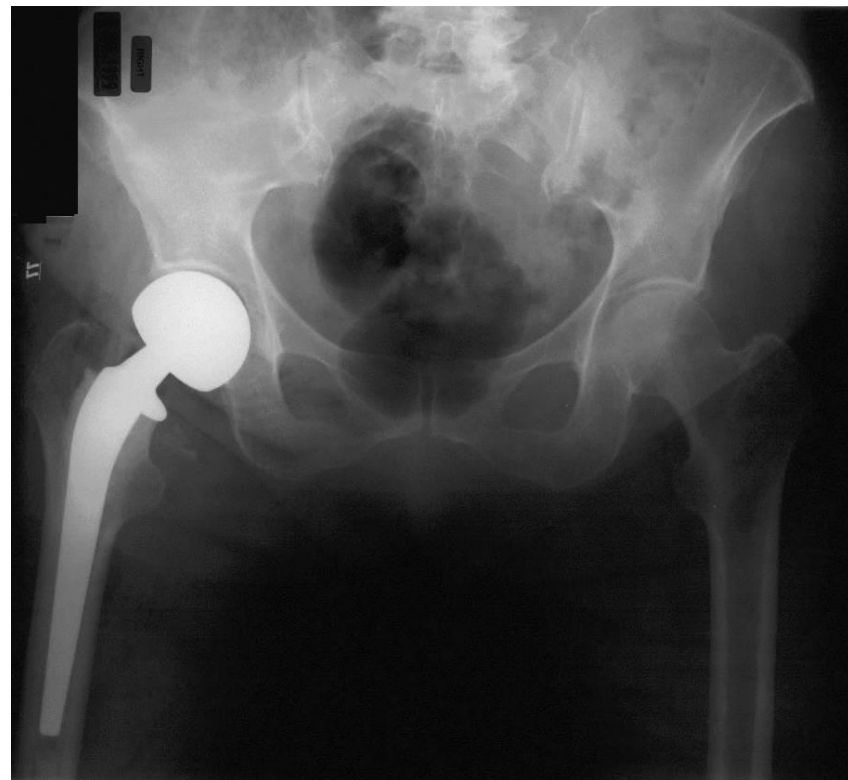


Displaced fracture of the neck of the femur

Proximal Femoral Fractures



Avascular necrosis of femoral head – flattening of weight-bearing upper portion against acetabulum and sclerosis (increased density – necrosis and calcification)

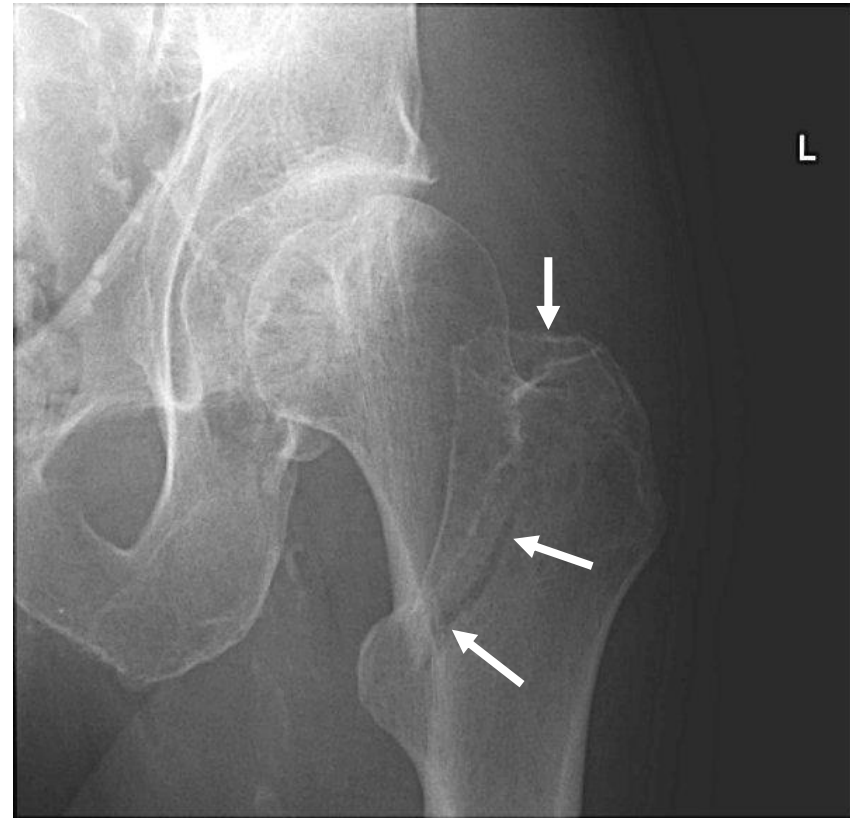


Partial hip replacement (**Hemiarthroplasty**) – possible treatment for fracture of the neck of the femur

Proximal Femoral Fractures

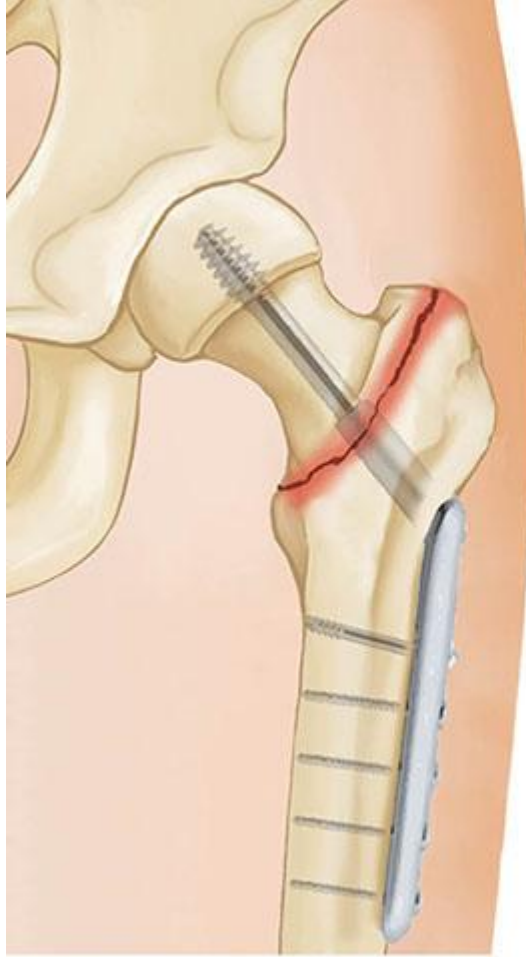
Extra-capsular – Intertrochanteric Fracture

- Fracture line runs between the two trochanters
- Does not affect blood supply to the femoral head
- Displaced extracapsular fractures (not shown in the image) can result in external rotation of the leg with some shortening



Nondisplaced intertrochanteric fracture

Proximal Femoral Fractures



Dynamic hip screw – possible treatment for
intertrochanteric fractures

Posterior Dislocation of the Hip Joint

- Posterior dislocations are most common
- The femoral head lies superior and lateral to the acetabulum
- **Ischiofemoral ligament is weak**
- The fibrous layer of the joint capsule ruptures inferiorly and posteriorly allowing the femoral head to pass through the tear in the capsule and over the posterior margin of the acetabulum onto the lateral surface of the ilium, **shortening and medially rotating the affected limb**
- What neurovascular injuries can occur due to this fracture?
 - **Sciatic Nerve**

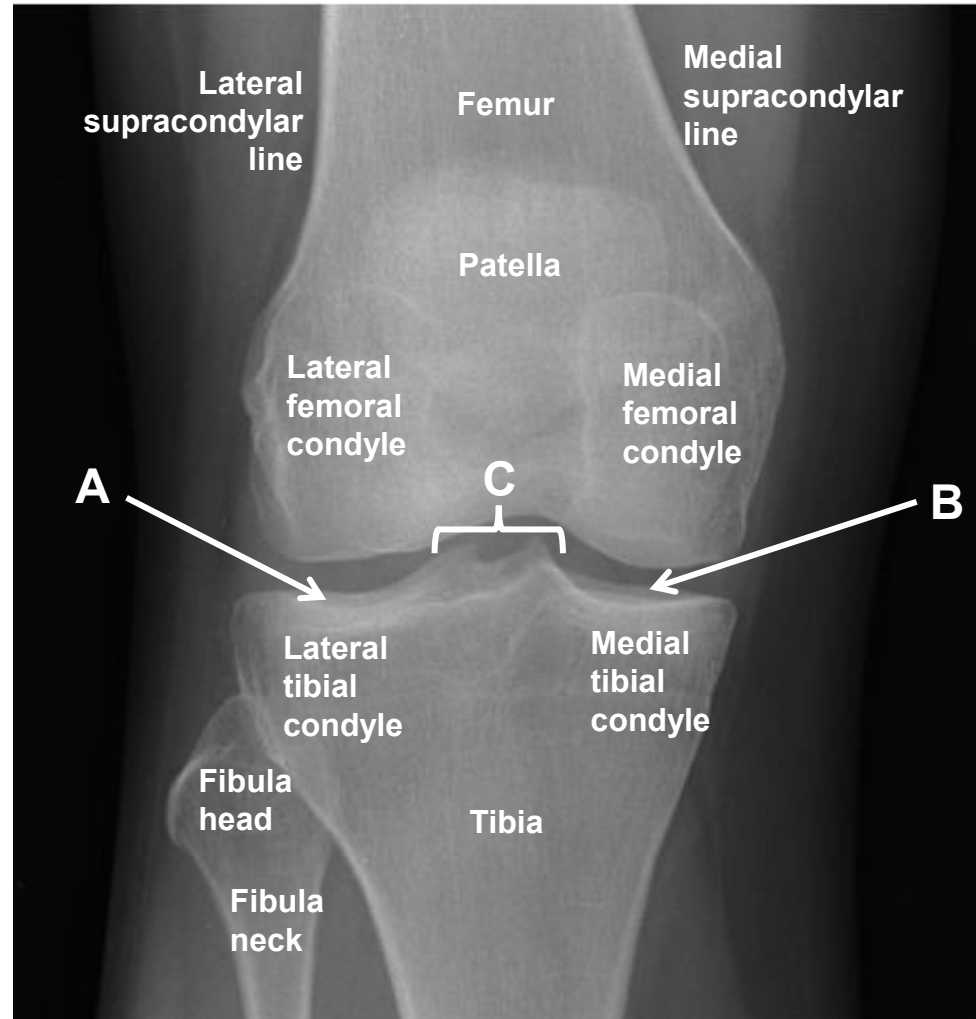


Case 3

A 30-year-old woman comes to the physician because of a 2-month history of intermittent left knee pain. The pain is mild to moderate in intensity. The patient describes the pain as sharp and dull. She cannot put weight on the left knee and is currently limping. Physical examination shows mild knee effusion. There is no pain with palpation along the medial or lateral border of the patella. Her active and passive range of motion is limited secondary to pain, but she is able to fully extend the knee and flex to 90 degrees. Collateral ligaments are intact to varus/valgus stress. The anterior drawer test is positive, and the posterior drawer is negative.

AP Radiograph of the Knee Joint

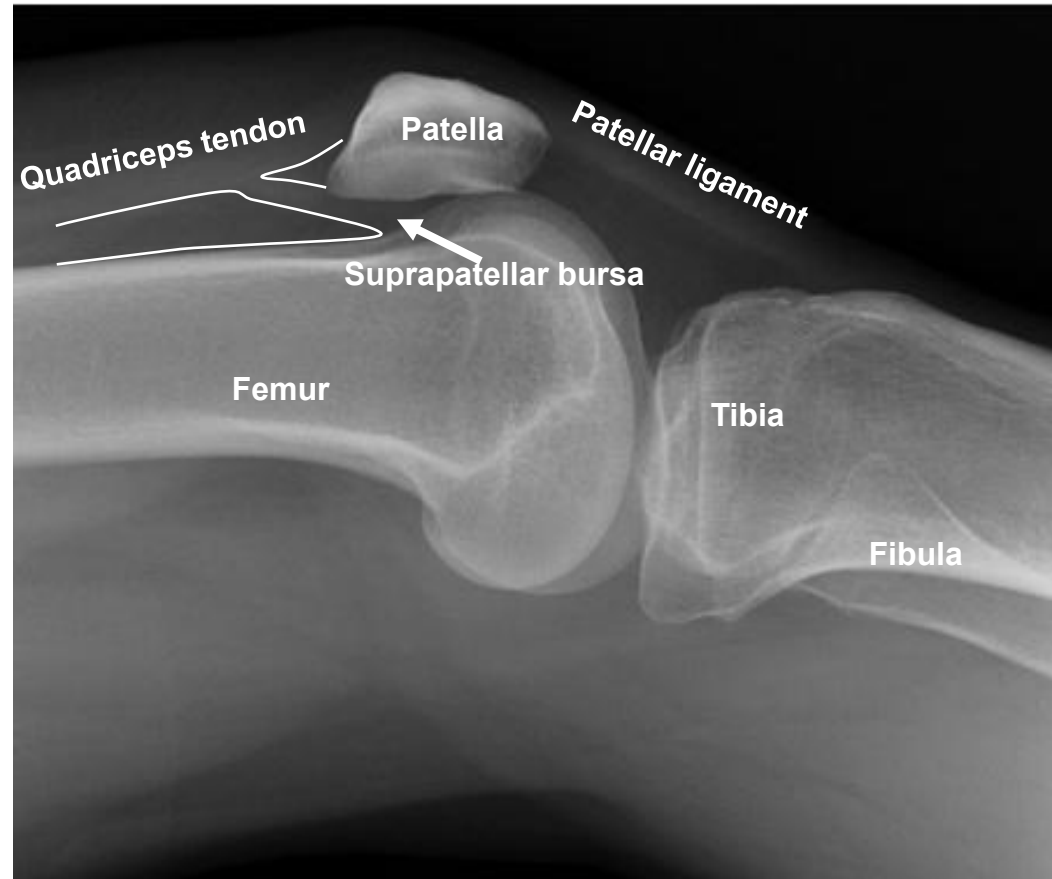
- Patella
- Femur
 - Lateral femoral condyle
 - Lateral supracondylar line
 - Medial femoral condyle
 - Medial supracondylar line
- Tibia
 - Lateral tibial condyle
 - A - lateral tibial plateau
 - Medial tibial condyle
 - B - medial tibial plateau
 - C - Intercondylar eminence with intercondylar tubercles
- Fibula
 - Fibula head
 - Fibula neck



Lateral Radiograph of the Knee Joint

Identify the anatomy

- quadriceps tendon
 - patella
 - patellar ligament
 - suprapatellar bursa
 - femur
 - tibia
 - fibula
- The horizontal beam lateral view is useful in assessing soft tissue as well as bones.
 - Quadriceps tendon and patellar ligament are visible.
 - The assessment of the suprapatellar bursa (an extension of the knee joint, not a true bursa) between the fat pads is important in trauma.

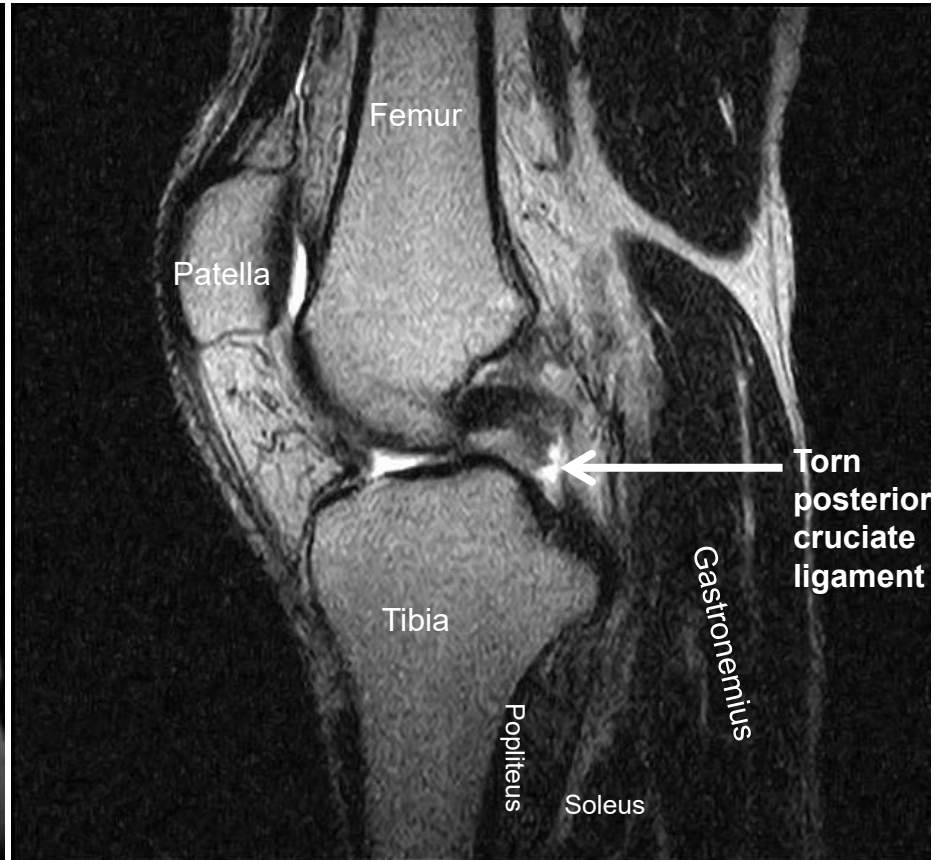


Posterior Cruciate Ligament

<https://radiopaedia.org/cases/posterior-cruciate-ligament-tear-2?lang=us>



MRI Knee - Sagittal T1 – Normal PCL



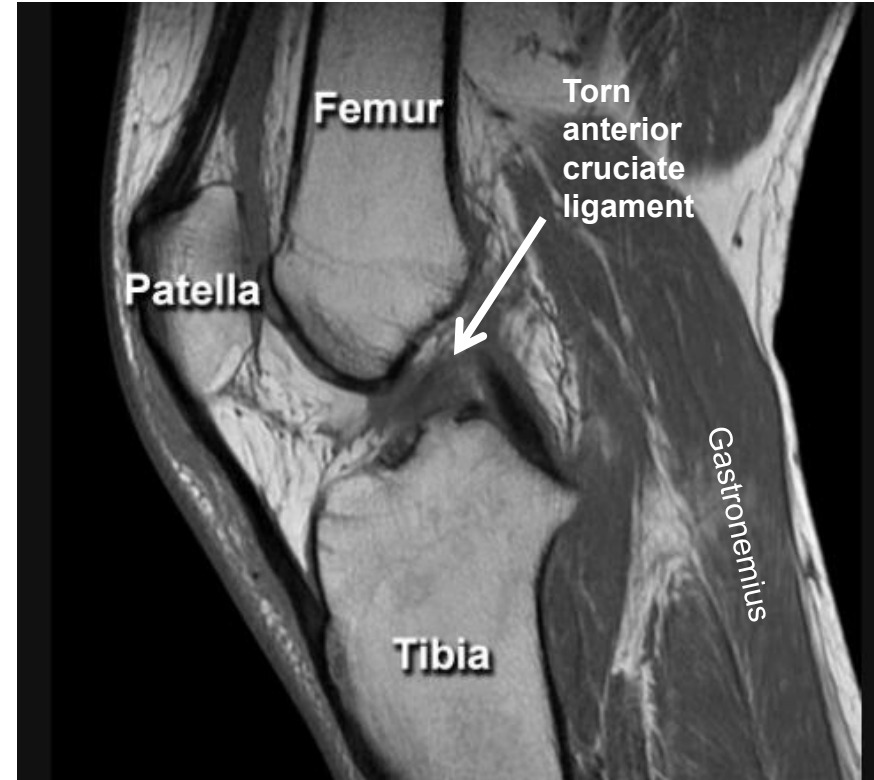
MRI Knee - Sagittal T2 – Torn PCL

Anterior Cruciate Ligament

<https://www.radiologymasterclass.co.uk/gallery/imaging-galleries/mri-gallery/mri-knee-acl-tear>

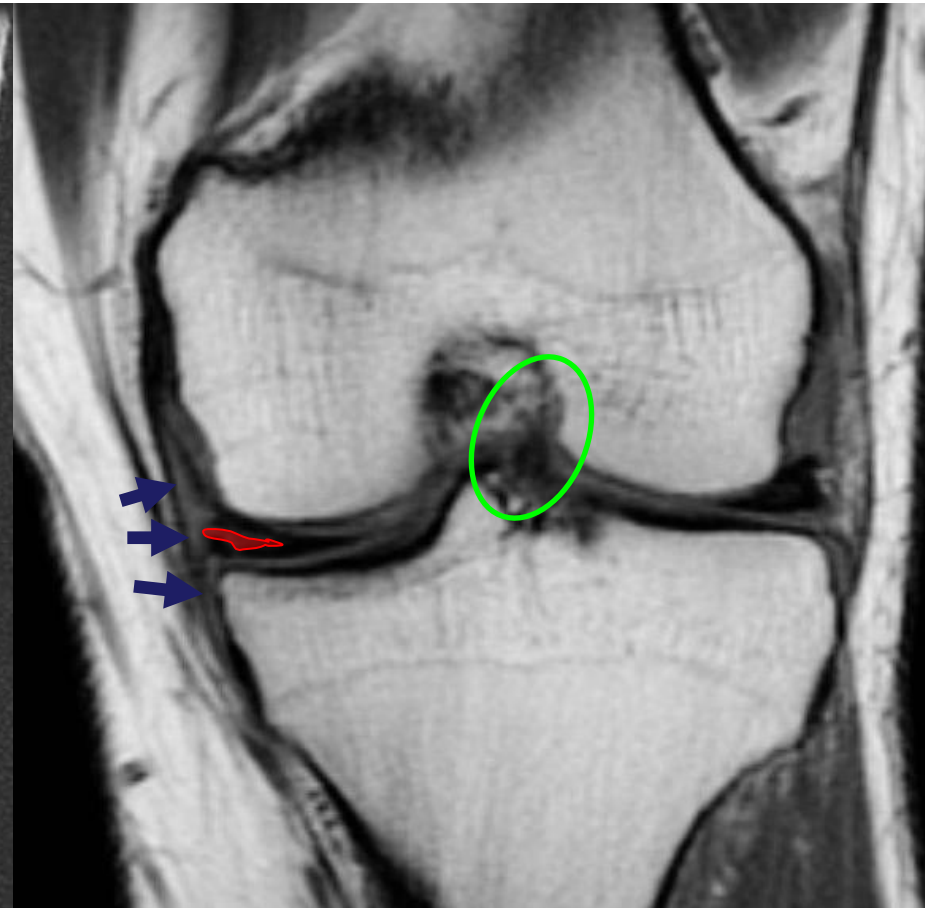
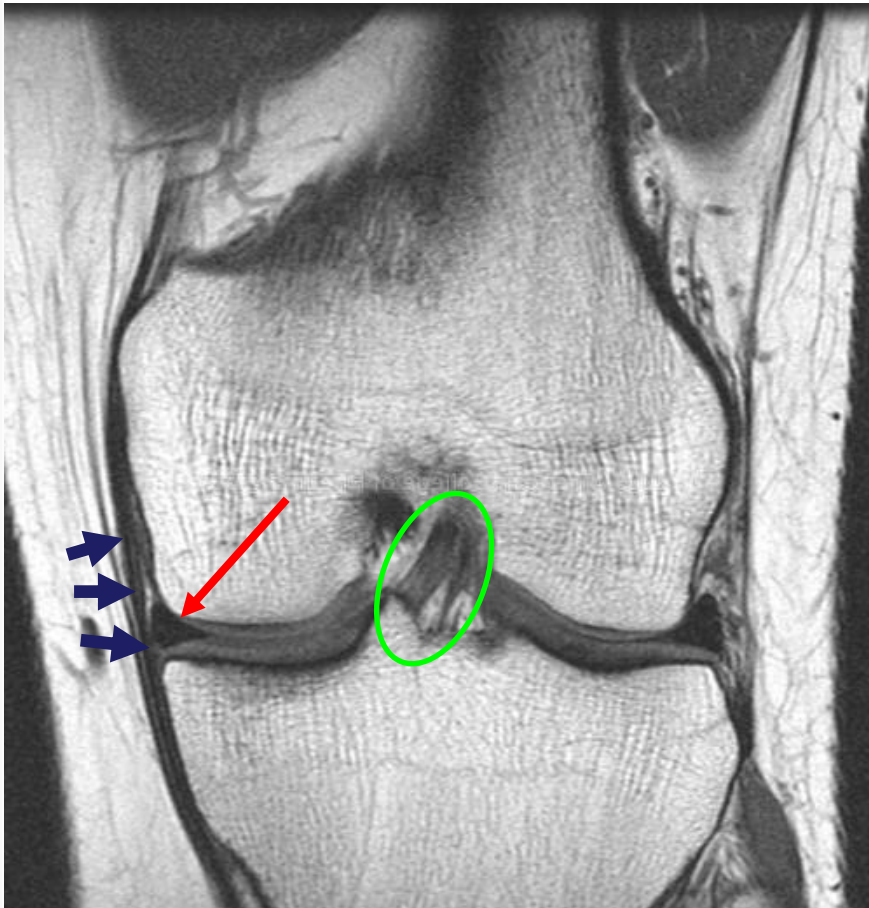


<https://www.radiologymasterclass.co.uk/gallery/imaging-galleries/mri-gallery/mri-knee-acl-tear>



MRI Knee - Sagittal T1 - Normal ACL MRI Knee - Sagittal T1 – Torn ACL

Unhappy Triad



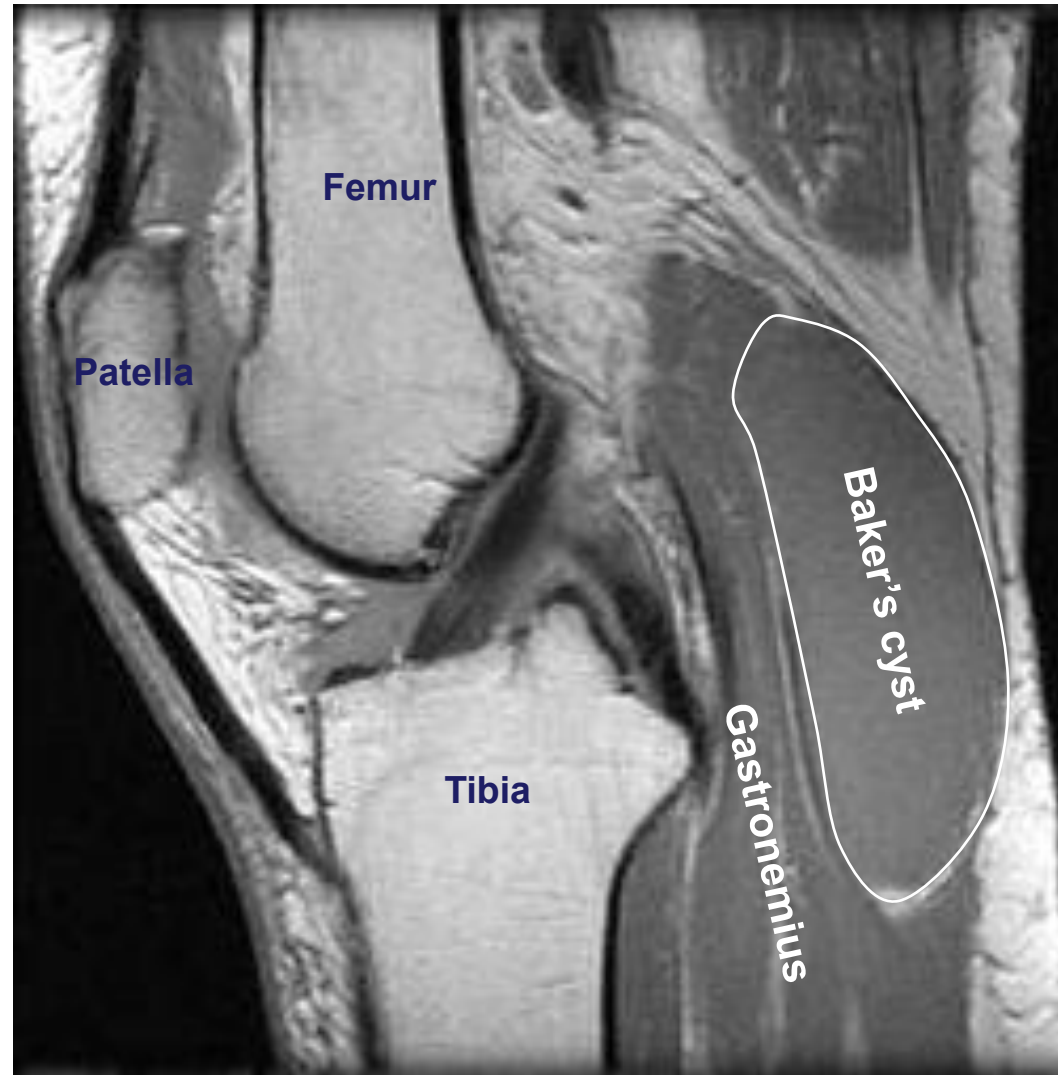
MRI Knee - Coronal T1 - Normal Knee

MRI Knee - Coronal T1- Unhappy Triad

Blue arrows: tibial collateral ligament
 Red arrow: medial meniscus
 Green line circle: anterior cruciate ligament

Baker's Cyst

- Collection of synovial fluid from the knee joint or a bursa forming a synovial-lined sac in the popliteal space
- Typically, the semimembranosus and gastrocnemius bursa
- They are usually located at or below the joint line.



Case 4

The day following his visit to the emergency department, while crossing a busy intersection, Gerald experiences excruciating in his right thigh which prevents him from walking any further. He is struck by car and suffers several injuries to the lower limb and pelvis.

What are some of the injuries that you are likely to see in this patient?

Distal Femoral Fractures

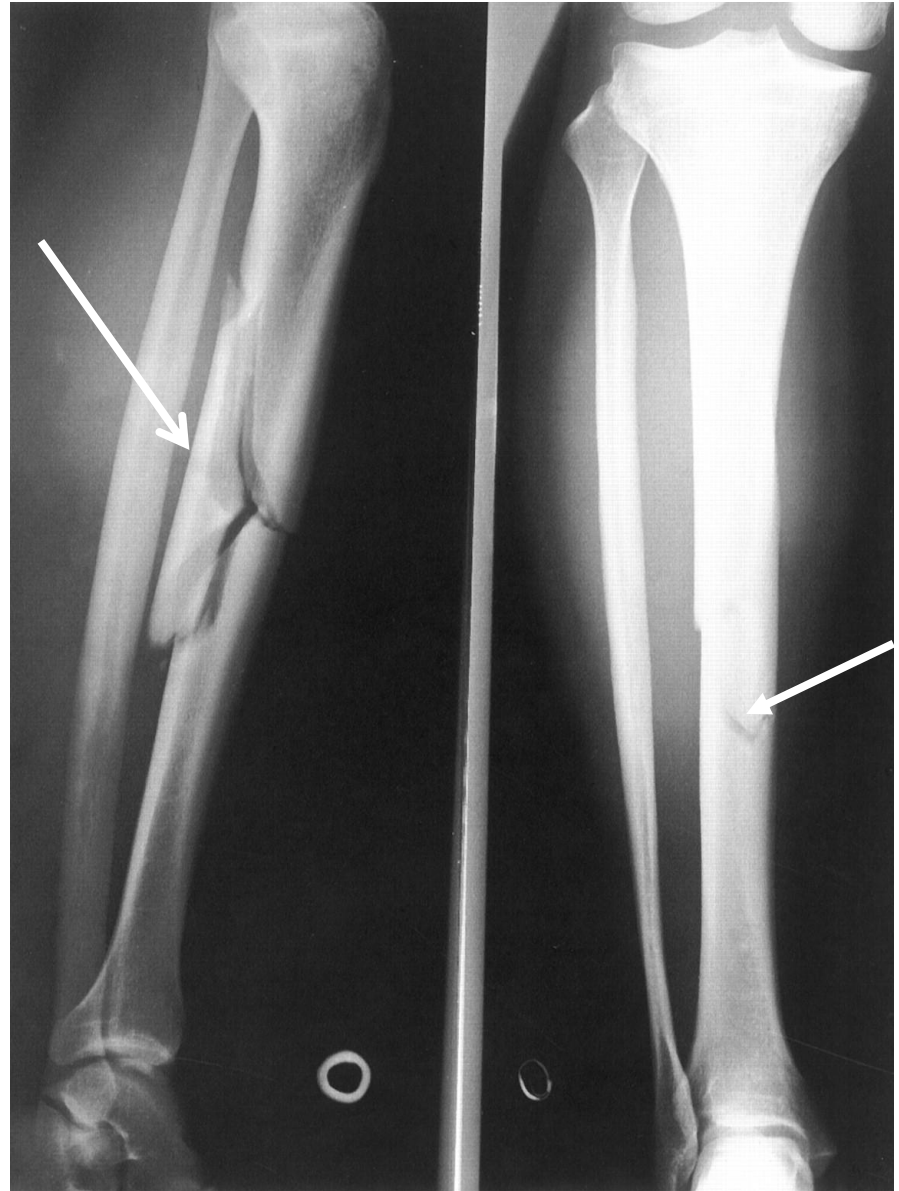
- Supracondylar fracture of the femur
- Seen in high velocity collisions
- Pain and inability to bear weight
- Risk of compartment syndrome (uncommon)
- What vessels can be damaged due to this injury?
 - Popliteal artery and vein



B

Tibial Fractures

- Comminuted fracture of the shaft of the tibia
- Can lead to compartment syndrome
- What structures can be damaged due to this injury?
 - Vessels and nerves of the anterior and/or posterior compartment



Compartment Syndrome

~45% of all cases of CS are caused by tibial fractures. Other causes include any long-bone fracture, burn injuries, vascular injury, crush injury, and tight casts or dressings.

Can have anterior, lateral or posterior compartment syndrome.

6 P's of acute limb ischemia:

- Pain
- Paresthesia's
- Pallor
- Paresis (weakness / paralysis)
- Pulselessness
- Poikilothermia (cold)

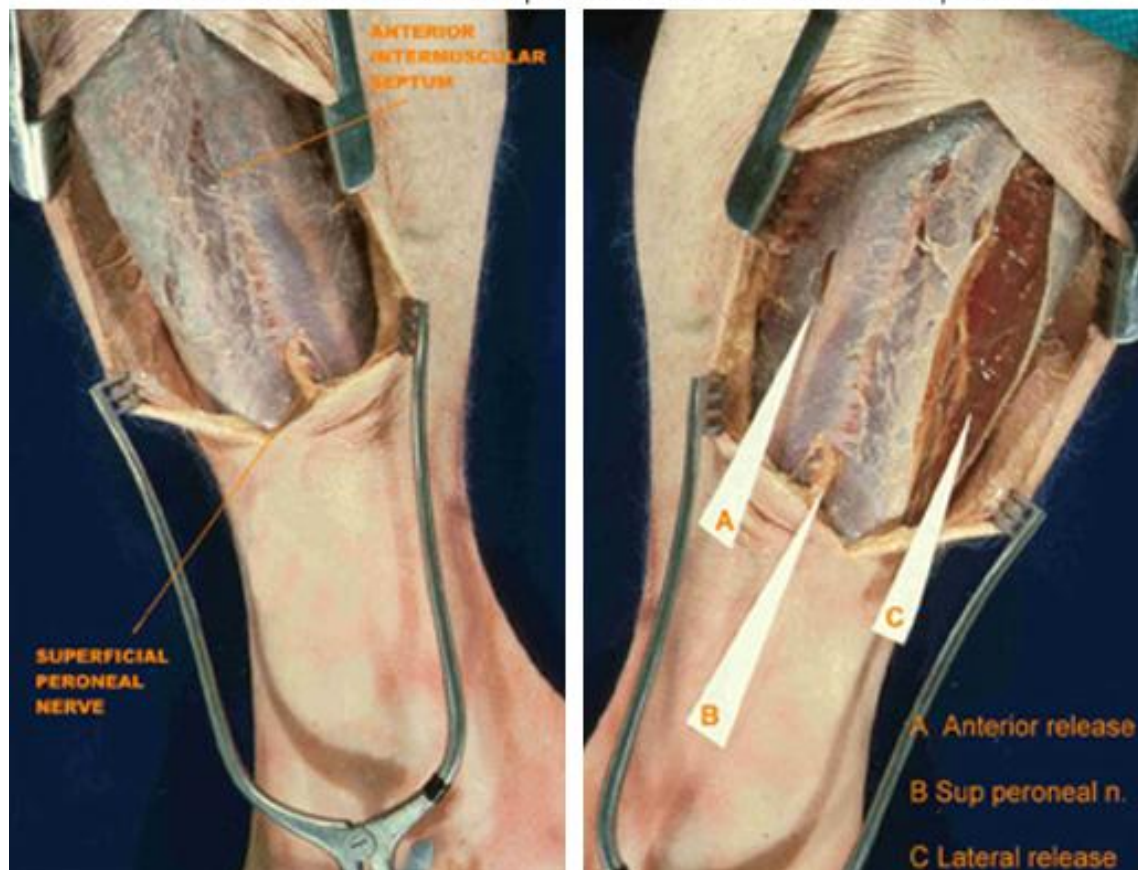


- This patient is being asked to point his feet to the ceiling.
- Which compartment is affected in this patient?
- Which nerve(s) would be affected in an anterior, posterior or lateral compartment syndrome?
- What functional deficits will the patient have?

Compartment Syndrome

The tissue pressure in a closed space rises above the perfusion pressure of microvasculature causing compromise of tissue circulation and function. Surgical treatment is release by incising the deep fascia with a fasciotomy

Anterolateral incision for decompression of anterior and lateral compartments



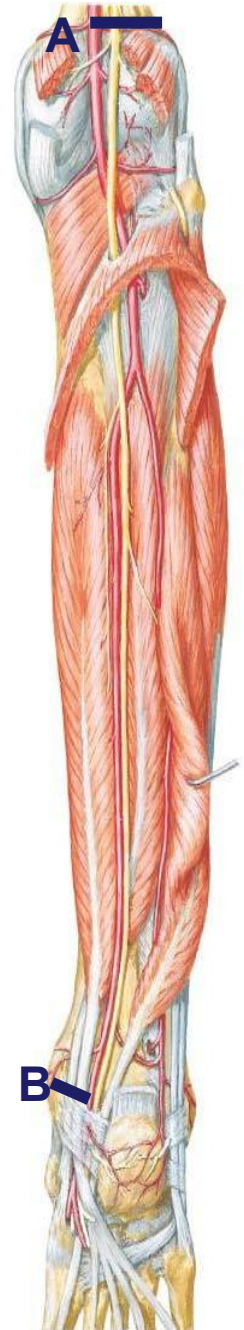
Summary of Tibial Nerve Injury

A: Tibial Nerve Injury in the Popliteal Fossa

- Paralysis of the muscles of the posterior compartment of the leg and the intrinsic muscles in the sole of the foot.
- Patients are unable to plantarflex the ankle and have decreased inversion of the ankle.
- Functions of the intrinsic joints of the feet will be lost (with the exception of extension)
- Loss of sensation on the posterolateral leg, heel and the sole of the foot

B: Tibial Nerve Injury in the Tarsal Tunnel

- Muscles of the posterior compartment of leg are spared, so patients can still plantarflex the ankle and inversion is not affected.
- Functions of the intrinsic joints of the feet will be lost (with the exception of extension)
- Loss of sensation on the sole of the foot



Summary of Common Fibular Nerve Injury

C: Common Fibular Nerve Injury at the Neck of Femur

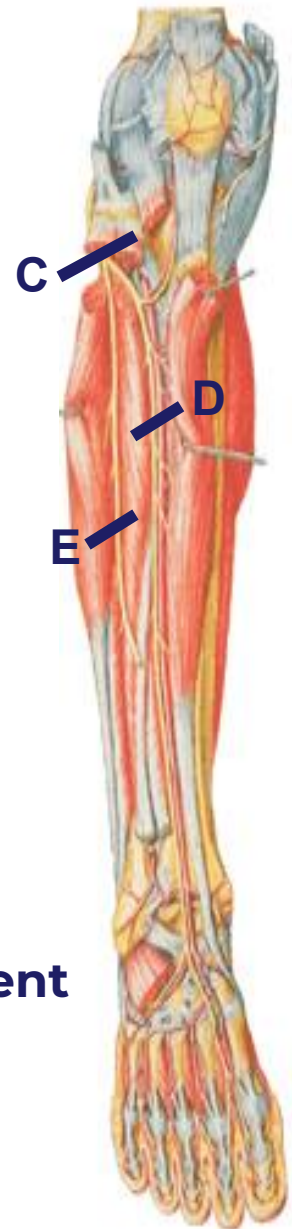
- Paralysis of muscles of the lateral and anterior compartments.
- Foot drop is the result of loss of dorsiflexion
- Lower limb becomes 'too long' so to compensate, the person will use a high stepping gait and/or swing out the leg
- There will also be decreased inversion of the foot
- Loss of sensation on the antero-inferolateral leg and entire dorsum of foot

D: Deep Fibular Nerve Injury in Anterior Compartment

- Paralysis of the muscles of the anterior compartment.
- Foot drop is the result of loss of dorsiflexion
- Lower limb becomes 'too long' so to compensate, the person will use a high stepping gait and/or swing out the leg
- There will also be decreased inversion of the foot
- Loss of sensation in the first web space

E: Superficial Fibular Nerve Injury in Lateral Compartment

- Paralysis of muscles of the lateral compartment.
- There will be loss of eversion of the foot
- Loss of sensation on the inferior-anterior leg and dorsum of foot except the first web space





CT Scan Series of the Lower Limb

